

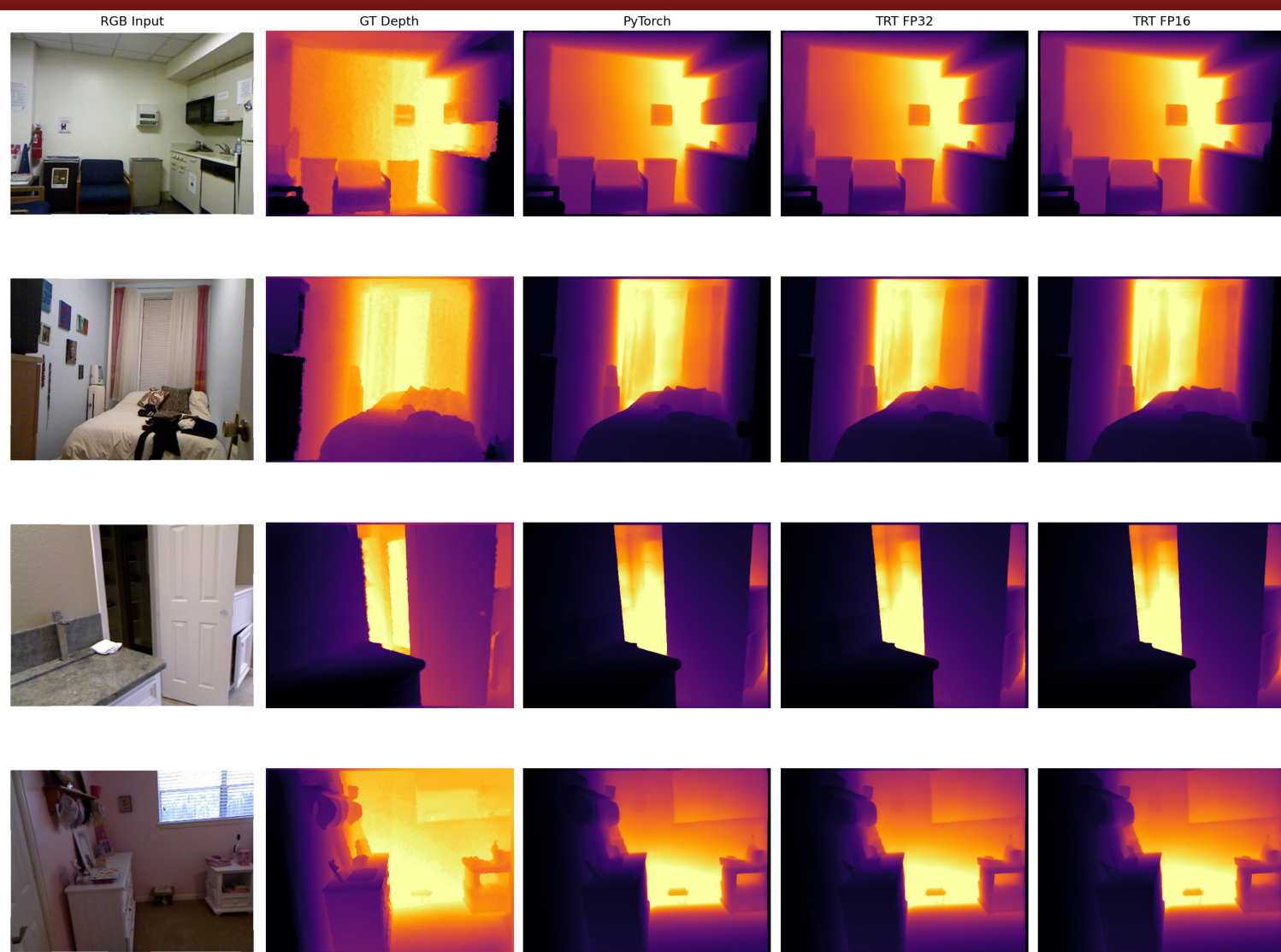


1. 서론

- ## 2. 실험 설계 및 방법

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- The diagram illustrates the conversion pipeline and benchmarking setup for Depth Anything V2 Variants. It is organized into three main horizontal sections.
- Depth Anything V2 Variants**
- This section contains three boxes representing the model variants:
- VT-S**: 24.8M params
 - VT-B**: 97.5M params
 - VT-L**: 335.3M params
- Arrows from these boxes point down to the Conversion Pipeline.
- Conversion Pipeline**
- This section shows the workflow for converting the models:
- PyTorch** (yellow box) is converted to **ONNX (opset 17)** (yellow box).
 - The ONNX model is then converted to a **Simplified ONNX** (yellow box).
 - The Simplified ONNX model is converted to the **TensorRT Engine** (yellow box).
 - The TensorRT Engine outputs two precision formats: **FP32** (green box) and **FP16** (orange box).
- An arrow from the FP32/FP16 outputs points down to the Benchmarking Setup.
- Benchmark on NVIDIA RTX 4070 Laptop GPU**
- This section contains three boxes detailing the benchmarking metrics:
- Latency (CUDA Events)**
 - mean, median
 - p95, p99
 - 200 iterations
 - Memory & Power (NVML)**
 - VRAM peak
 - Avg power (W)
 - Accuracy (WU Depth V2)**
 - AbsRel, RMSE
 - S_1, S_2, S_3
 - 654 samples

3. 실험 결과: 종합 성능 및 트레이드오프



- 자연시간-정확도 트레이드오프
 - FP16 변환으로 모든 모델 4.4~4.8배 속도 향상
 - ViT-S/FP16: 233 FPS로 30 FPS 기준 8배 → 실시간 엣지 배포 최적
 - ViT-B/FP16: 81.8 FPS로 실시간 가능
 - ViT-L/FP16: 26.7 FPS로 신시간 한계
- 정확도 보존
 - FP32-FP16 정확도 차이 AbsRel 0.0002 이하 → 사실상 손실 없는 변환
 - 시각적으로도 PyTorch/FP32/FP16 depth map 식별 불가
 - LayerNorm 분포가 FP16에 강건/모델 크기에 따라 정확도 단조 증가

4. 실험 결과: 레이어 단위 분석

- | Category | ViT-S/FP32 | ViT-S/FP16 | ViT-B/FP32 | ViT-B/FP16 | ViT-L/FP32 | ViT-L/FP16 |
|-----------|------------|------------|------------|------------|------------|------------|
| Attention | 55.4 | 24.0 | 48.5 | 21.9 | 45.5 | 20.3 |
| FFN | 15.5 | 25.3 | 21.5 | 36.6 | 24.5 | 37.5 |
| LayerNorm | 2.2 | 4.8 | 1.7 | 3.0 | 1.5 | 3.2 |
| DPT Head | 17.4 | 33.4 | 15.6 | 23.9 | 15.5 | 23.5 |

5. 결론 및 향후 연구

- ## 6. 참고 문헌

[9] R. Ranftl, A. Bochkovskiy, and V. Koltun, “Vision Transformers for Dense Prediction,” Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2021, pp. 12179–12188.